

Module 8—Exercise 1—Solution file—Jude Ataba

Exercise 1: Merging Customer Files

Course Text: *Programming Logic & Design*, 10th Edition

Programming Exercise: #2 (page 236)

Problem Description

Geraldine's Landscaping Service and Gerard's Landscaping Service are merging their businesses and must merge their customer files into a single file. Each customer file contains:

- Customer Number
- Last Name
- Address
- Property Area in Square Feet

Both files are already sorted in ascending order by customer number, and there are no duplicate customer numbers between the two files.

Solution Overview

The solution uses a sequential file merge algorithm. One record from each input file is read into memory and compared by customer number. The record with the smaller customer number is written to the output file. The program then reads the next record from the file that provided the record that was written. This continues until one file reaches end-of-file. Any remaining records from the other file are then written to the output file. All files are closed at the end of processing to ensure data integrity.

This approach follows standard file-handling practices:

- Files are declared using appropriate file types (InputFile/OutputFile)
- Records are read into memory before processing
- Output is written to a new file
- Files are closed after processing

- **Testing Scenarios and Results**

Input Files Used

- Geraldines Businesses.csv
- Gerald's Businesses.csv

Test Scenario 1: Both files contain multiple records

- **Expected Result:** Output file contains all records sorted by customer number
- **Result:** Passed

Test Scenario 2: Verify sorted order

- **Expected Result:** Customer numbers are in ascending order
- **Result:** Passed

Test Scenario 3: Verify record count

- **Expected Result:** Output record count equals total records from both files
- **Result:** Passed

Test Scenario 4: Verify no duplicate customer numbers

- **Expected Result:** No repeated customer numbers appear in output
- **Result:** Passed

Proof of Output (Merged File Evidence)

Record Count Verification

- Geraldines Businesses.csv contains **5** records
- Gerald's Businesses.csv contains **5** records
- Merged output contains **10** records

This confirms that no records were lost during the merge.

Merged Output File (Full Output)

CustomerNumber,LastName,Address,PropertyAreaSqFt

1598,Jones,Jeffersonville,1500

2578,Smith,Evansville,2500

4891,Smith,South Bend,1500

4978,Miller,Bloomington,9800

5144,Johnson,Terre Haute,3500

6548,Robertson,Indianapolis,1000

8949,Garcia,Fort Wayne,2500

9841,Brown,Gary,3000

9853,Erickson,Clarksville,1750

9856,Jacobs,Columbus,1900

Output Validation

- Customer numbers are in ascending order
- Output includes all records from both input files
- No duplicate customer numbers appear
- Output file layout matches the input file layout

Conclusion

All testing scenarios passed successfully. The merged output file contains all customer records in correct customer-number order and meets the assignment requirements.